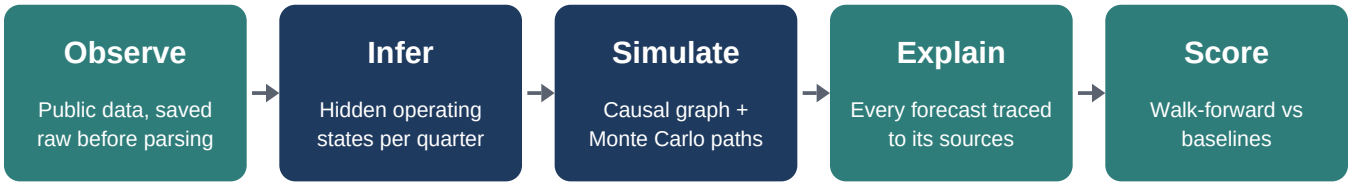
Chui Twin: status report

Longaeva AI Hackathon, Stage 2 | Benedict Ngetich | October 1, 2026 | For Doug Rogers

Summary

- **Chui Twin is now a system rather than a single-company model.** One engine turns any public company whose results come from observable operating drivers into a runnable operating model. **16 twins are live across 5 industries**, and 14 of them were generated from the ticker alone, with no hand-written configuration, in about 1 to 3 minutes each.
- **Every twin is scored the same way.** Each one is refit at every historical earnings release using only data that was public at that moment, then compared with what was reported. Across the portfolio the twin beats a "same as last quarter" forecast on **20 of 33** KPI series and a "same quarter last year" forecast on **31 of 33**.
- **Every number is traceable** to a filing accession number and exhibit, or to a saved raw copy of the source, down to the sentence a value was extracted from.
- **The results are reported honestly, including where the twin loses.** Chipotle, the deep example, currently only ties the naive forecast, and Delta's own guidance still beats the twin on revenue. Both findings shape the plan below.

How it works



Public data is treated as noisy sensors of a company's hidden operating states: demand, capacity, price, input cost, labor, sentiment and expansion. The engine estimates those states each quarter, links them to reported KPIs through a causal graph with stated priors, and simulates thousands of paths forward. Each forecast comes with a distribution, a driver breakdown down to the individual signal, and scenario analysis. For example, a sentiment shock the size of a food-safety scare cuts Chipotle's simulated next-quarter comps by about 10 points (in the actual 2015 E. coli quarter, comps fell 14.6%).

Why it scales

Layer	What is shared	What a new company or industry needs
Engine	Estimation, simulation, attribution and scoring. An automated test fails if any company, industry or signal name appears in engine code.	Nothing.
Data adapters	8 sources (SEC EDGAR and XBRL, FRED, BLS, TSA, Wikimedia, GDELT news tone, public sitemaps with Internet Archive history). TSA serves every airline; FRED, BLS and GDELT serve every company.	Usually nothing. New data sources are added once and reused.
Industry packs	One short YAML file per industry: which public signals inform which states, and default KPI structure. Five exist: restaurants, airlines, hotels, homebuilders, retail.	A new industry needs one pack file, not new code.
Twin factory	Ticker to SEC record to industry pack. The company's Wikipedia page and news query are accepted only if verified by its ticker. KPIs come from XBRL, which every SEC filer publishes in the same format.	Just the ticker.

Evidence: the portfolio scorecard

Each twin is tested walk-forward over 2017 to mid-2026 (26 to 31 quarters per company) on two GAAP KPIs from XBRL, revenue growth and operating margin, plus company-specific KPIs for the hand-built twins. COVID quarters are excluded from scoring (travel companies through 2022). "Beats" means a lower CRPS, a standard score for probabilistic forecasts that rewards both accuracy and honest uncertainty. The 14 factory twins use industry defaults with no per-company tuning.

Industry	Twins	KPI series	Beats last quarter	Beats same quarter last year	Example
Airlines	DAL, UAL, AAL, LUV	8	8	7	UAL operating margin: mean error 2.2 pts vs 4.6 naive
Homebuilders	DHI, LEN, PHM	4	3	4	PHM revenue growth: CRPS 33% better than naive
Hotels	MAR, HLT	4	2	3	HLT operating margin: CRPS 30% better than naive
General retail	TGT, WMT	4	2	4	WMT operating margin: mean error 0.43 pts vs 0.58
Restaurants	CMG, SBUX, MCD, DRI, TXRH	13	5	13	SBUX operating margin: CRPS 63% better than naive
Total	16 twins	33	20	31	

Full per-company results, including the KPI series where the twin loses, are in docs/portfolio_scorecard.md in the repository and are regenerated by one command.

The Chipotle deep example

- **Ground truth:** 59 quarters of comparable sales and restaurant-level margin (2011 to 2026) extracted automatically from Chipotle's earnings releases. Each value keeps the sentence it came from, and spot checks match the tables in Chipotle's own releases.
- **Company-specific signals:** US restaurant count (2,975 in mid-2022 to 4,166 today) and open job postings, counted daily from Chipotle's public sitemaps, with history rebuilt from Internet Archive captures. Also news tone

(GDELT), web attention, and macro restaurant sales, menu inflation, wages and food costs.

- **Result so far:** on comps the twin ties the naive forecast (mean error 3.8 vs 3.7 points) and is well ahead of the seasonal baseline. Our reading is that free macro data carries little Chipotle-specific information about next quarter's comps, which is why the plan below focuses on expectations and company-level data.
- **Live test:** a sealed forecast of Chipotle's Q3 2026 comps (its SHA-256 hash published beforehand) will be checked against the October 28 report.

The three bars you suggested

Bar	Status
Seasonal naive (same quarter last year)	Built into every scorecard. The twin beats it on 31 of 33 KPI series.
Management guidance	Extracted automatically from earnings releases (Delta: 12 quarters, 2023 to 2026, each with its source sentence). **Guidance currently beats the twin** on Delta revenue growth (mean error 3.3 vs 5.2 points). Next step: use guidance as the anchor and model the surprise relative to it.
Consensus	Not yet included. Consensus estimates are licensed data, so we are looking for a legal source (for example, figures published in press coverage, entered with citations).

Next steps

When	Milestone
By Oct 9 (Stage 2)	**Expectation gap:** use guidance as the starting point and forecast the surprise from the signals, for any company that guides. **Dashboard:** each twin's forecast, drivers traced to sources, scenario sliders and the portfolio scorecard. Consensus baseline if a legal source works.
By Oct 21 (finals)	LLM onboarding: a model reads the 10-K to choose company-specific KPIs (comps, RevPAR, home deliveries) and drafts the extraction rules, with every value keeping its source text. Third deep twin. A shock library (food safety, fuel spikes, strikes) with decay profiles measured from history and reused across companies.
Oct 28	Chipotle reports Q3. The sealed forecast resolves and gets scored publicly.

Data and compliance

All sources are public and listed in a provenance file, with each source's terms and access method. Sites' robots.txt rules are respected (for example, Chipotle's careers site blocks its job-search path, which we never request). Requests are rate-limited, and nothing is scraped from behind a login or an access key; for that reason Chipotle's menu prices are not collected. Backtests use only data whose publication time precedes the forecast date. One known limitation is that macro series are stored in their latest revised form, which slightly flatters history; the fix (archived original releases) is planned.

Ask

A 30-minute Zoom early next week (October 5 or 6), whenever suits you. I would value your view on two questions: (1) is a cross-industry portfolio scorecard the right way to prove scalability to the judges, or would you emphasize something else; and (2) do you know a legitimate source for consensus estimates we could use?